

TrustSyn

Biologically Informed Drug-Combination Prediction and Trust-Aware Prioritization

OncoNova Team · Novartis Discoverathon 2026 · Challenge 1: Drug Combination Synergy Prediction

Dataset: NCI-ALMANAC

Team members: Dr. Sushant · Anoushka Sreeram · Konuri Kala Sahithi

Source-of-truth statement

This report is generated from the frozen artifact package **Complete_model.zip**, received 16 August 2026, which is treated as the single integrated source of truth for all reported results. The earlier standalone D-MPNN V2-C model-development documentation is retained and cited for architecture, dataset, and methodology detail where it does not conflict with the frozen package. Any earlier-stage numeric result that differs from the frozen figures is explicitly flagged rather than merged or silently overwritten.

Identity convention: OncoNova = team name. TrustSyn = model/system name. This distinction is maintained throughout the document.

1. Title and Team

Field	Value
Project / system name	TrustSyn
Team name	OncoNova
Competition	Novartis Discoverathon 2026 — Challenge 1
Challenge focus	Drug-combination synergy prediction using NCI-ALMANAC
Team members	Dr. Sushant, Sahithi, Anoushka

Primary prediction target combo_score (NCI-ALMANAC)

Source-of-truth artifact Complete_model.zip, received 16 August 2026

2. Executive Summary

TrustSyn is OncoNova's integrated framework for predicting anti-cancer drug-combination activity on the NCI-ALMANAC **combo_score** target. The system combines biologically informed tabular features, a CatBoost branch, a directed message-passing neural network (D-MPNN V2-C), split-specific stacking, a trust and uncertainty layer, and explainability artifacts.

The core methodological decision is to evaluate every model under four canonical settings that deliberately separate interpolation from extrapolation:

- **RANDOM** — ordinary IID performance.
- **COLD_COMBINATION** — unseen drug pairings.
- **COLD_CELL_LINE** — unseen biological context.
- **COLD_DRUG** — unseen drug chemistry.

This split design, and the explicit refusal to let any one setting stand in for the others, is the throughline of the entire submission.

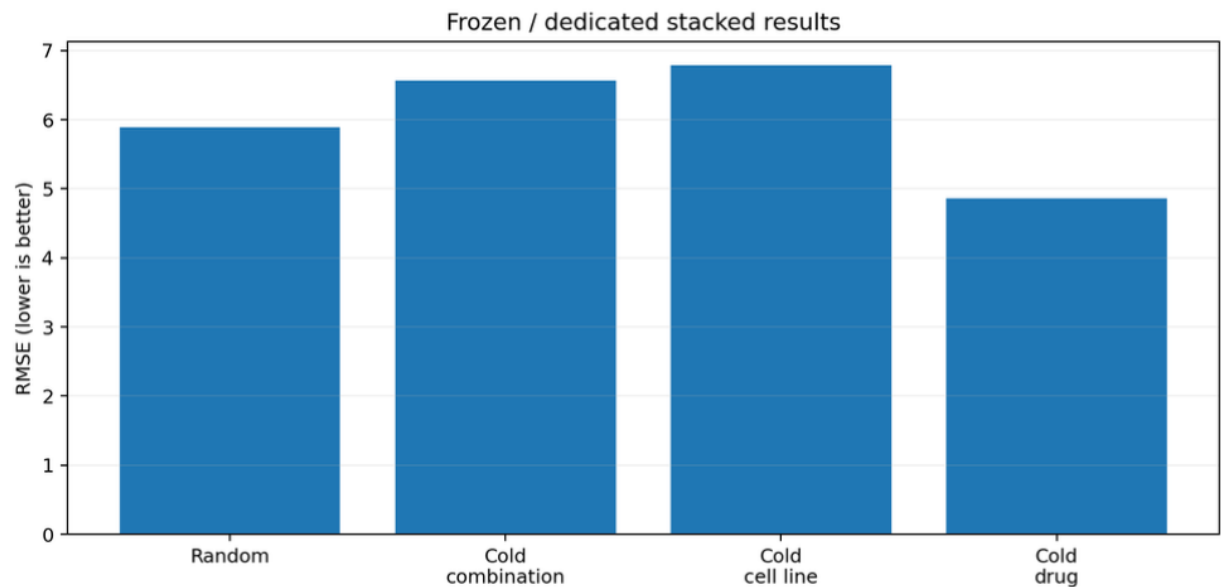
Main Results — Frozen Stacked Artifacts

Split	Selected Meta-Model	RMSE	R ²
RANDOM	Ridge	5.888	0.294
COLD_COMBINATION	XGBoost	6.568	0.129
COLD_CELL_LINE	XGBoost	6.789	0.323
COLD_DRUG	Dedicated stacked artifact	4.858	0.351

Key Findings

- Stacking improves and stabilizes predictions relative to any single base model, but no architecture is uniformly best. CatBoost has the lowest individual MAE on every split, while D-MPNN contributes a structurally different, molecular-graph-based signal that is complementary rather than superior.

Figure: Frozen / Dedicated Stacked Results



RMSE (lower is better) of the frozen split-specific stacked artifacts across the four canonical splits, corresponding to the Main Results table: RANDOM (Ridge, 5.888), COLD_COMBINATION (XGBoost, 6.568), COLD_CELL_LINE (XGBoost, 6.789), and COLD_DRUG (dedicated stacked artifact, 4.858).

- Generalization degrades under distribution shift in a non-uniform way. COLD_CELL_LINE shows the largest MAE gap versus RANDOM for the D-MPNN component, while COLD_DRUG shows the sharpest collapse in correlation and ranking quality even though its raw MAE gap looks smaller.
- The trust layer, comprising conformal intervals, novelty scoring, and trust categories, is empirically well calibrated on RANDOM, COLD_COMBINATION, and COLD_CELL_LINE, covering 88,753 rows with approximately 88.5–90.8% empirical coverage against a 90% nominal target. It is explicitly out of scope for COLD_DRUG.
- Novelty score is not a consistently reliable error predictor across splits. Its correlation with absolute error ranges from approximately +0.047 to −0.187. It is therefore reported as a supporting signal rather than a validated confidence measure.

Key Limitations

- Classification-style metrics — Macro-F1, balanced accuracy, AUPRC, AUROC, and MCC — for a defined synergy-hit threshold were not part of the supplied evaluation artifacts and are not reported.
 - A complete 1/5/10/25/100% data-efficiency curve exists only for COLD_COMBINATION. The other three splits currently have only the 100% point.
 - No clinical, in-vitro, or wet-lab validation has been performed. All results are computational and are explicitly not claims of biological mechanism.
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3. Problem and Challenge Objective

Combining two or more anti-cancer drugs can produce effects that are stronger, weaker, or qualitatively different from what each drug does alone. Identifying combinations that act synergistically — where the joint effect exceeds what independent action would predict — is a central strategy in oncology drug development because synergistic combinations can improve efficacy, reduce required dosing, and help overcome resistance.

Screening all possible drug-pair-by-cell-line combinations experimentally is infeasible at scale. This motivates predictive models that can prioritize which combinations are worth testing.

Challenge 1 of the Novartis Discoverathon 2026 frames this as a supervised prediction problem over the NCI-ALMANAC dataset, a large public screen of paired anti-cancer drug combinations tested across a panel of NCI-60 cancer cell lines.

The prediction target throughout this submission is **combo_score**, the ALMANAC combination-activity score for a given drug pair, cell line, and concentration pairing.

The Discoverathon evaluation deliberately goes beyond a single random train/test split. The central scientific question is not simply:

"Can a model fit ALMANAC?"

Instead, the more important question is:

"Can a model generalize to genuinely unseen combinations, unseen cell lines, and unseen drugs, and can it do so in a way that is sample-efficient rather than requiring exhaustive re-screening?"

TrustSyn is designed around this emphasis. Every architecture, split, and evaluation metric exists to test extrapolation, not just interpolation, and to be explicit about where extrapolation fails.

4. Dataset

TrustSyn is built on a frozen, canonical TrustSyn ALMANAC master dataset derived from NCI-ALMANAC.

Dataset summary

- **File:** `almanac_final_59cell_104drug.csv`
- **Observations:** 294,073
- **Cancer cell lines:** 59
- **Unique drugs:** 104
- **Original columns:** 9

Primary Fields

Field	Description
drug_A / drug_B	Identifiers for the two drugs in the combination
CELLNAME	Cell-line identifier as recorded in ALMANAC
tissue	Tissue of origin for the cell line
combo_score	Prediction target — ALMANAC combination-activity score
CONC1 / CONC2	Tested concentrations of drug A and drug B
cellminer_cellline_id	Mapping key to the CellMiner biological feature resource

nci_almanac_cellname Cell-line name used by the original NCI-ALMANAC release

For compatibility with the D-MPNN training and evaluation pipeline, `combo_score` is additionally exposed under the internal column name `target`; the underlying values are unchanged.

The master dataset itself was not modified for D-MPNN training. The same frozen file underlies the tabular CatBoost/LightGBM and molecular-graph D-MPNN branches of TrustSyn.

5. Data Curation and Quality Control

The supplied artifact package documents curation at the level of the frozen master file rather than as a fully itemized cleaning log.

What is verified from the source documentation:

- The ALMANAC master dataset, `almanac_final_59cell_104drug.csv`, is treated as frozen.
- It was not modified for D-MPNN training.
- The same master file is shared across the CatBoost, LightGBM, and D-MPNN branches, avoiding branch-specific data drift.
- All 59 canonical ALMANAC cell lines were verified against their CellMiner representations before biological features were constructed.
- Cell lines without a resolvable CellMiner mapping were not silently incorporated into the biological feature pipeline.
- Drug-pair identity is treated as order-independent downstream. The modeling design assumes that `drug_A/drug_B` labeling is not itself biologically meaningful.

Limitation

The supplied documentation does not include a complete step-by-step data-cleaning log, such as explicit duplicate-row counts, missing-value imputation rules, or detailed drug-name canonicalization tables. This report therefore includes only what is verifiable from the frozen-dataset and CellMiner-verification documentation rather than inferring undocumented procedures.

6. Leakage Prevention and Split Design

Leakage-safe evaluation is the organizing principle of TrustSyn.

Every reported result is tagged with the split from which it came. No split's training data is allowed to contain information that would let a model circumvent the specific generalization question represented by that split.

The canonical TrustSyn split definitions were held fixed for both the tabular and D-MPNN branches. Splits were not regenerated independently for each model.

Split	Test Size	Purpose / Leakage Rule
RANDOM	29,408	Ordinary IID evaluation. Observations are randomly held out while preserving the overall data distribution. Drug pairs, cell lines, and individual drugs may reappear between train and test.
COLD_COMBINATION	29,558	Tests generalization to drug combinations absent from training. The specific drug pairing in the test set does not appear in training, although individual drugs may appear in other combinations.
COLD_CELL_LINE	29,787	Tests generalization to cell-line contexts absent from training. Held-out cell lines do not appear in the training partition.
COLD_DRUG	3,132	Strongest stress test. Held-out drugs are absent from training entirely, requiring extrapolation to molecular chemistry never seen during training.

The COLD_DRUG test partition is markedly smaller than the other three splits — 3,132 versus approximately 29,400–29,800 rows.

This occurs because removing entire drugs from training also removes every combination involving those drugs. This is considerably more restrictive than holding out combinations or cell lines alone.

The smaller test set is itself important because COLD_DRUG metrics should be interpreted with greater uncertainty than the other splits.

7. Feature Engineering

7.1 Tabular Branch — CatBoost / LightGBM

The tabular models consume engineered features spanning chemical, target-based, network, pathway, and cell-biological information.

The globally important features identified by CatBoost SHAP analysis include:

Feature	Family
union_target_count	Combined drug-target set size
Tanimoto_similarity	Chemical fingerprint similarity
target_count_A / target_count_B	Individual drug-target set sizes
STRING_available	Availability of STRING protein-interaction information
CellMiner_PC2 / CellMiner_PC24	Principal components of CellMiner biological features
KEGG_overlap	Pathway overlap between drug targets
target_overlap_B / target_jaccard	Shared versus distinct target-set structure

Together, these features represent three major engineered families:

1. **Chemical similarity** — Tanimoto fingerprint similarity.
2. **Target/pathway biology** — target counts, target overlap, STRING, and KEGG.
3. **Cell-line molecular state** — CellMiner principal components.

7.2 Molecular-Graph Branch — D-MPNN

Rather than relying primarily on precomputed fingerprints or descriptors, the D-MPNN branch represents each drug directly as a molecular graph:

$$G = (V, E)$$

where V represents atom nodes and E represents chemical-bond edges.

The final configuration uses node features of dimension 7 and edge features of dimension 6.

Biological context is provided by the same CellMiner resource used by the tabular branch.

CellMiner Feature Group	Source	Approx. Feature Count
RNA expression	<code>cellminer_rna_features_59cell.csv</code>	~20,202
Copy-number variation	<code>cellminer_cnv_features_59cell.csv</code>	~19,282
Mutation	<code>cellminer_mutation_features_59cell.csv</code>	~9,307
Protein	<code>cellminer_protein_features_59cell.csv</code>	~94

Because this biological feature space is extremely high-dimensional, it is compressed using PCA to approximately 50 components before being supplied to the D-MPNN cell-line encoder.

8. Model Development

TrustSyn evaluates three principal model families before combining their predictions.

CatBoost

CatBoost is the primary tabular gradient-boosting branch. It operates on engineered chemical, target, network, pathway, and cell-biological features.

It is the strongest single model by MAE across all four splits and therefore anchors the meta-model comparison for stacking.

LightGBM

LightGBM is a secondary tabular gradient-boosting baseline. Its role is to establish whether the molecular-graph representation provides complementary information beyond multiple tabular approaches.

D-MPNN V2-C

D-MPNN V2-C is a directed message-passing neural network operating directly on drug molecular graphs combined with compressed CellMiner biological context.

Its purpose is not simply to outperform CatBoost. Instead, it provides a structurally different representation of the drug pair that can potentially contribute complementary information to an ensemble.

These three components feed a split-specific stacking layer.

No single base model is presented as the final TrustSyn output.

9. D-MPNN Architecture

D-MPNN V2-C consists of three principal components:

1. A shared drug encoder applied independently to each drug.
2. A symmetric pair-representation layer.
3. A cell-line encoder supplying biological context.

These representations are then combined by a fusion network and passed to the final regression head.

9.1 Molecular Representation

Each drug is represented as a molecular graph:

$$G = (V, E)$$

where atoms are nodes and chemical bonds are edges.

The model therefore operates directly on molecular structure rather than treating each molecule exclusively as a fixed-length engineered vector.

9.2 Directed Message Passing

The core molecular encoder is a Directed Message Passing Neural Network.

Unlike standard graph convolution, which propagates information symmetrically between neighboring nodes, D-MPNN passes messages over directed bonds.

The directed-bond states are iteratively updated across message-passing steps. These states are then aggregated into atom-level representations and ultimately into a whole-molecule embedding.

The goal is to allow the model to learn local chemical environments and progressively construct a molecular representation rather than relying entirely on manually engineered descriptors.

9.3 Shared Drug Encoder

The same D-MPNN encoder, with shared learned weights, is applied to drug A and drug B.

There is no separate network for each drug position.

This design:

- reduces the number of model parameters;
- ensures that the same molecule receives the same representation regardless of whether it appears as drug_A or drug_B;
- supports chemically meaningful symmetry between the two drug inputs.

9.4 Pair Representation

Given drug embeddings $\mathbf{h}_A$ and $\mathbf{h}_B$, TrustSyn constructs a symmetric pair representation using:

- element-wise sum;
- element-wise product;
- element-wise absolute difference.

These operations capture complementary information:

- **Sum** captures combined molecular representation.
- **Product** captures feature-wise interaction.
- **Absolute difference** captures similarity or dissimilarity.

The use of symmetric operations is deliberate. Drug-combination prediction should not depend on which compound happens to be entered as drug_A versus drug_B.

9.5 Cell-Line Representation and Fusion

Molecular information alone is insufficient because the same drug pair can behave differently in different biological contexts.

TrustSyn therefore compresses CellMiner RNA, CNV, mutation, and protein measurements into approximately 50 PCA components.

These components are passed through the cell-line encoder to create a biological embedding.

The cell embedding is then fused with the drug-pair representation.

The fusion network can therefore learn interactions such as:

- whether a particular drug-pair representation behaves differently in a specific cellular context;
- whether molecular similarity is predictive only for certain biological states;
- whether the same combination behaves differently across cell-line backgrounds.

9.6 Final Configuration

The evaluated architecture is D-MPNN V2-C.

Hyperparameter	Value
Node feature dimension	7
Edge feature dimension	6
Cell embedding dimension	200
Hidden dimension	128
Depth	2
Dropout	0.1
Learning rate	0.001
Batch size	256
Random seed	42

Each canonical split has its own dedicated checkpoint directory:

```
FINAL_V2_C/{RANDOM, COLD_COMBINATION, COLD_CELL_LINE,  
COLD_DRUG}/checkpoints/
```

D-MPNN was therefore trained independently for each split rather than trained once and re-evaluated across multiple held-out partitions.

10. Baseline Models

CatBoost and LightGBM were evaluated as tabular baselines against which D-MPNN could be assessed.

The purpose was to establish:

1. whether learned molecular representations add predictive value over engineered features;
2. whether D-MPNN contributes complementary information;
3. which base models are suitable for stacking.

D-MPNN vs. CatBoost — Individual Model MAE

Split	D-MPNN MAE	CatBoost MAE	D-MPNN – CatBoost
RANDOM	4.064	3.209	+0.855
COLD_COMBINATION	4.335	3.967	+0.368
COLD_CELL_LINE	4.765	3.206	+1.558
COLD_DRUG	4.481	3.967	+0.514

CatBoost has lower individual MAE than D-MPNN on all four splits.

D-MPNN is therefore not presented as outperforming CatBoost in raw predictive accuracy.

Its value lies in providing a structurally different molecular representation that may contribute complementary information to the ensemble.

D-MPNN vs. LightGBM — Individual Model MAE

Split	D-MPNN MAE	LightGBM MAE	D-MPNN – LightGBM
RANDOM	4.064	4.588	-0.525
COLD_COMBINATION	4.335	4.512	-0.177
COLD_CELL_LINE	4.765	5.223	-0.458
COLD_DRUG	4.481	4.181	+0.300

D-MPNN improves on LightGBM on RANDOM, COLD_COMBINATION, and COLD_CELL_LINE.

LightGBM remains slightly better on COLD_DRUG.

Provenance Note

These comparisons come from the standalone D-MPNN model-development documentation and compare individual models only. They are not the final stacked ensemble results. The separate CatBoost evaluation package contains additional optimized CatBoost results, but those optimized values are not substituted into the frozen stacking inputs.

11. Ensemble / Stacking

TrustSyn's reported main results are produced by a split-specific stacking layer.

The base-model predictions from CatBoost, LightGBM, and D-MPNN are combined using a meta-model selected for the relevant split.

Split	Selected Stacking Artifact	RMSE	R ²
RANDOM	Ridge meta-model	5.888	0.294
COLD_COMBINATION	XGBoost meta-model	6.568	0.129
COLD_CELL_LINE	XGBoost meta-model	6.789	0.323
COLD_DRUG	Dedicated stacked artifact	4.858	0.351

Ridge regression is used as the selected meta-model for RANDOM.

XGBoost is used for COLD_COMBINATION and COLD_CELL_LINE.

COLD_DRUG uses a dedicated, separately produced stacked artifact rather than the shared three-split stacking pipeline.

This distinction is important because the trust layer does not cover COLD_DRUG.

The separation avoids implying a common validation scope where one does not exist.

12. Trust Layer

The TrustSyn trust layer spans **88,753 rows** across:

- RANDOM;
- COLD_COMBINATION;
- COLD_CELL_LINE.

It combines:

- CatBoost predictions;
- D-MPNN predictions;
- ensemble agreement;
- uncertainty flags;
- conformal prediction intervals;
- novelty scoring;
- derived trust categories.

The trust layer is a reliability-supporting layer built around the prediction artifacts.

Scope

The trust layer explicitly does not cover COLD_DRUG. Any trust or confidence statement in this report applies only to RANDOM, COLD_COMBINATION, and COLD_CELL_LINE unless explicitly stated otherwise.

Conformal Interval Coverage

Nominal coverage: **90%**

Split	Rows	Empirical Coverage	Mean Interval Width
RANDOM	29,408	90.69%	19.21
COLD_COMBINATION	29,558	90.82%	19.21
COLD_CELL_LINE	29,787	88.49%	19.21

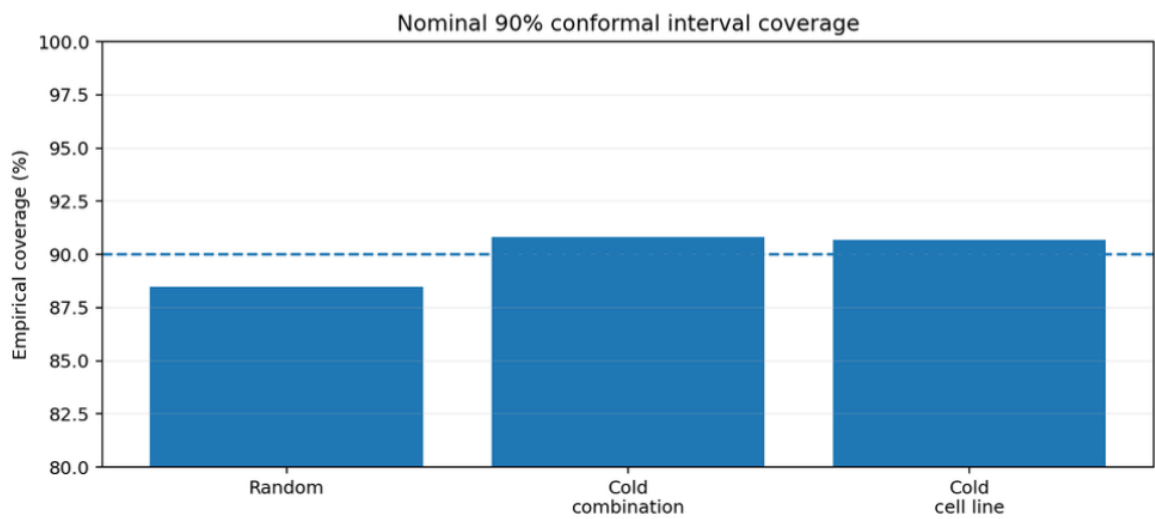
Empirical coverage remains close to the 90% nominal target across all three covered splits.

COLD_CELL_LINE is slightly under-covered at 88.49%, consistent with the split having a larger underlying generalization gap.

Trust-Category Reliability

Split	High-Trust MAE	Medium-Trust MAE	Low-Trust MAE
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Figure: Nominal 90% Conformal Interval Coverage



Empirical coverage of the 90% nominal conformal intervals for RANDOM (90.69%), COLD_COMBINATION (90.82%), and COLD_CELL_LINE (88.49%). COLD_CELL_LINE is mildly under-covered, consistent with its larger generalization gap. COLD_DRUG is outside the trust layer's validated scope and is not shown.

RANDOM	3.72	3.93	5.06
COLD_COMBINATION	3.95	4.06	4.80
COLD_CELL_LINE	4.20	4.23	5.40

Observed error increases from high-trust to low-trust predictions on every covered split.

This is the intended behavior of the trust layer: predictions classified as low-trust are empirically less accurate.

13. Explainability

Explainability is provided through global SHAP feature attribution on the CatBoost branch.

The strongest global features include:

1. `union_target_count`
2. `Tanimoto_similarity`
3. `target_count_A`
4. `STRING_available`
5. `CellMiner_PC2`
6. `target_count_B`
7. `KEGG_overlap`

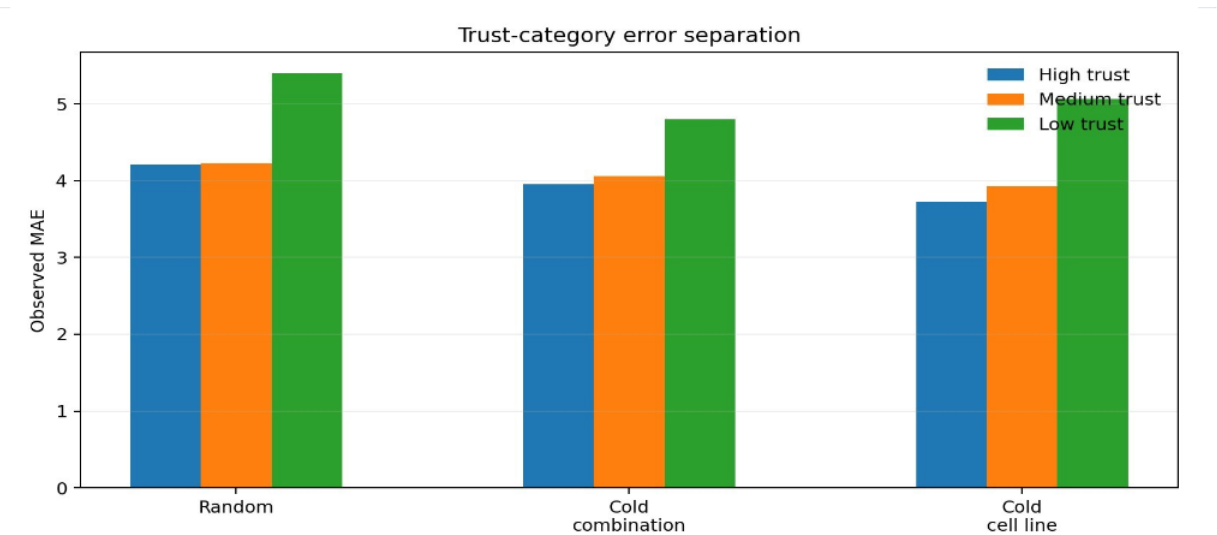
These features span:

- target-set structure;
- chemical similarity;
- network information;
- pathway information;
- cell-line molecular state.

Important Scientific Qualification

SHAP attribution describes what the model relied upon statistically when producing predictions. It is not evidence of biological mechanism and does not establish that a feature causally drives drug synergy. SHAP should therefore be interpreted as model explainability rather than mechanistic biology.

Figure: Trust-Category Error Separation



Observed MAE by trust category (High / Medium / Low) across RANDOM, COLD_COMBINATION, and COLD_CELL_LINE. Error increases monotonically from high-trust to low-trust predictions on every covered split, indicating the trust categories perform genuine error-separation work.

14. Evaluation Protocol

14.1 Regression Metrics

MAE

Mean Absolute Error is the primary evaluation metric.

Lower values are better.

RMSE

Root Mean Squared Error penalizes large prediction errors more strongly than MAE.

Lower values are better.

Pearson Correlation

Pearson correlation measures linear association between predicted and observed `combo_score`.

Higher values are better.

Spearman Correlation

Spearman correlation measures rank-order agreement between predicted and observed `combo_score`.

Higher values are better.

R^2

R^2 measures the proportion of variance explained by the predictions.

Higher values are generally better.

14.2 Ranking Metrics

Precision@K

Measures the proportion of the top-K predicted combinations that are actually high-performing according to the evaluation target.

Recall@100

Measures how many of the truly relevant combinations are recovered within the top 100 predictions.

nDCG@100

Normalized Discounted Cumulative Gain evaluates whether relevant combinations appear near the top of the ranked list.

Enrichment@100

Measures how enriched the top-100 predictions are for relevant combinations relative to their baseline prevalence.

Higher values are better for all ranking metrics.

During D-MPNN training, mean squared error was used as the optimization loss, while MAE was used as the principal validation and early-stopping criterion.

Thus, the best checkpoint was selected using the same primary metric used for final accuracy reporting.

15. Main Results

The headline TrustSyn results are the frozen split-specific stacked artifacts described in Section 11.

These are the numbers OncoNova considers the primary output of the project.

Split	Selected Stacking Artifact	RMSE	R ²
RANDOM	Ridge meta-model	5.888	0.294
COLD_COMBINATION	XGBoost meta-model	6.568	0.129
COLD_CELL_LINE	XGBoost meta-model	6.789	0.323
COLD_DRUG	Dedicated stacked artifact	4.858	0.351

Reporting Qualification

The frozen stacking output reports RMSE and R^2 but does not contain a complete stacked-ensemble MAE, Pearson/Spearman, or ranking table in the supplied artifacts.

Where this report reports MAE, Pearson, Spearman, or ranking metrics elsewhere, those values refer to the individual D-MPNN model and are explicitly labeled as such rather than being presented as measurements of the stacked ensemble.

16. Ranking Results

Ranking evidence is available for the individual D-MPNN V2-C model.

Split	Precision@10 0	Recall@100	nDCG@100	Enrichment@100
RANDOM	0.38	0.38	0.611	111.75
COLD_COMBINATION	0.15	0.15	0.434	44.34
COLD_CELL_LINE	0.29	0.29	0.518	86.38
COLD_DRUG	0.05	0.05	-0.083	1.57

Ranking quality follows the overall regression pattern:

- strongest on RANDOM;
- weaker on COLD_COMBINATION;
- degraded but still informative on COLD_CELL_LINE;
- severely degraded on COLD_DRUG.

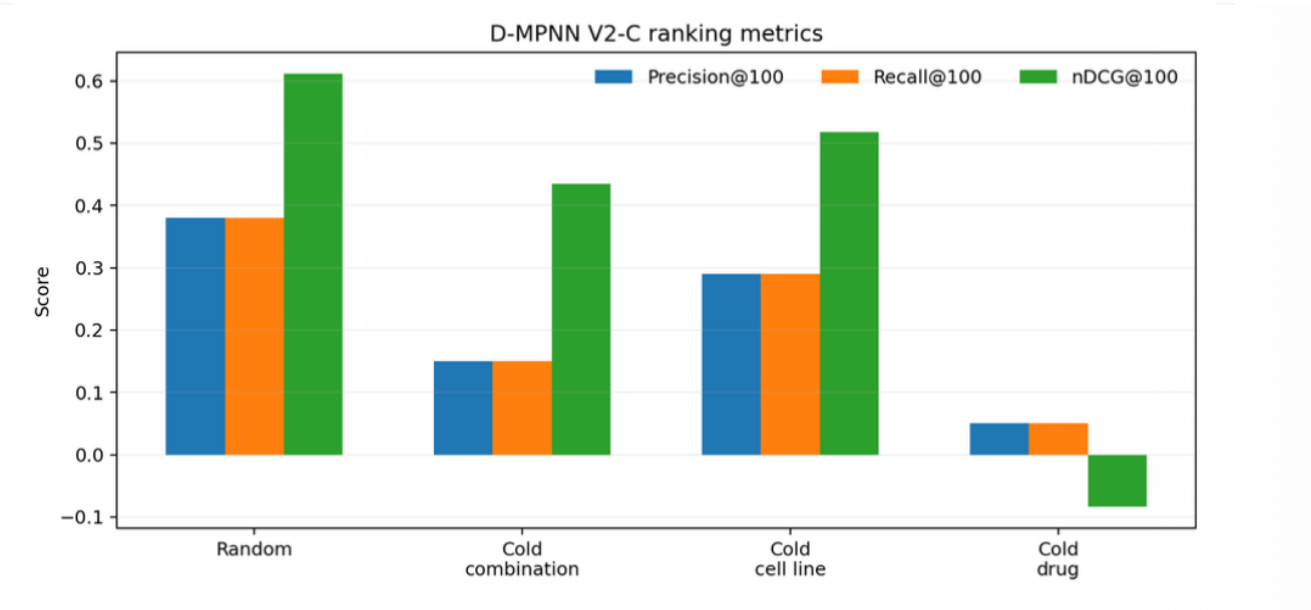
The COLD_DRUG Enrichment@100 value of 1.57 is close to baseline, while RANDOM reaches 111.75.

Ranking Coverage Limitation

No stacked-ensemble-level ranking table was included in the supplied artifacts. The ranking values above therefore describe the individual D-MPNN model.

The standalone D-MPNN documentation also contains an earlier ranking computation with materially different values. Because those values conflict with the frozen evaluation package, they are treated as superseded and are not merged into the final ranking table.

Figure: D-MPNN V2-C Ranking Metrics



Precision@100, Recall@100, and nDCG@100 for the individual D-MPNN V2-C model across the four canonical splits. Ranking quality is strongest on RANDOM and severely degraded on COLD_DRUG, where nDCG@100 becomes negative.

17. Classification Results

Not Available

Classification-style metrics for a defined synergy-hit threshold — including Macro-F1, balanced accuracy, AUPRC, AUROC, and MCC — were not included in the supplied evaluation artifacts.

No definitive synergy-hit threshold was established in the reviewed documentation.

Rather than deriving potentially misleading classification values from regression outputs, these metrics are explicitly marked as unavailable.

18. Data-Efficiency Analysis

Data efficiency evaluates how test performance changes as the fraction of available training data decreases.

A complete 1/5/10/25/100% curve is available for the D-MPNN model on COLD_COMBINATION.

Training Fraction	Training Rows	Test MAE	Test RMSE
1%	2,346	4.545	7.029
5%	11,590	4.627	7.113
10%	23,405	4.661	7.151
25%	58,699	4.675	7.150
100%	235,050	4.335	6.613

The curve is relatively flat.

MAE changes only modestly from 4.545 at 1% of the training data to approximately 4.675 at 25%, before improving to 4.335 at 100%.

This suggests that COLD_COMBINATION performance reaches a relatively stable regime with a small fraction of the available training data.

However, the 100% result remains the canonical final result.

Coverage Limitation

Only COLD_COMBINATION currently has the complete 1/5/10/25/100% curve.

RANDOM, COLD_CELL_LINE, and COLD_DRUG currently have only their 100% data point available.

The evaluation workbook therefore marks the data-efficiency component as **PARTIAL**.

19. Generalization Analysis

Generalization gap is defined as:

$$\text{MAE_gap} = \text{MAE_cold} - \text{MAE_random}$$

Using the individual D-MPNN RANDOM MAE of 4.064 as the reference:

Split	Random MAE	Split MAE	MAE Gap	Relative Gap
RANDOM	4.064	4.064	0.000	0.0%
COLD_COMBINATION	4.064	4.335	0.271	6.7%
COLD_CELL_LINE	4.064	4.765	0.701	17.2%
COLD_DRUG	4.064	4.481	0.417	10.3%

By raw MAE gap, COLD_CELL_LINE is the hardest split.

However, COLD_DRUG presents a more severe failure when correlation and ranking are considered.

For example:

- Pearson decreases to approximately 0.077.
- Spearman decreases to approximately 0.062.
- nDCG@100 becomes negative.

- Enrichment@100 approaches baseline.

Therefore, MAE gap alone understates the difficulty of COLD_DRUG.

A model can maintain a moderate average absolute error while still producing an almost uninformative ranking.

20. Baseline Comparison

The D-MPNN evaluation workbook includes a qualitative RMSE comparison among:

- D-MPNN V2-C;
- CatBoost V2;
- LightGBM.

The overall pattern is consistent with the detailed MAE comparisons:

- CatBoost is the strongest individual model on every split.
- D-MPNN improves over LightGBM on three of four splits.
- All three models weaken substantially under COLD_DRUG.
- No single individual architecture dominates every evaluation dimension.

The comparison deliberately concerns individual models rather than the final TrustSyn stacked output.

The separate CatBoost evaluation package contains additional optimized CatBoost results, but these are not substituted into the frozen stacking inputs.

21. Robustness / Model Agreement

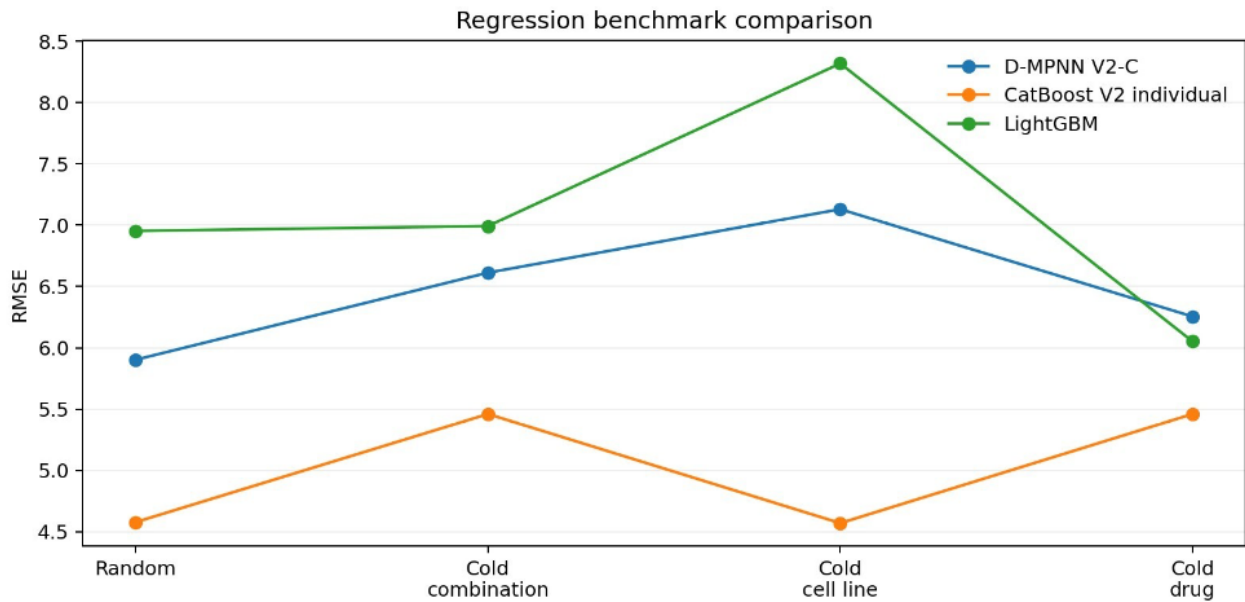
Robustness is evaluated across three complementary axes:

1. predictive accuracy;
2. correlation;
3. ranking quality.

A model can be weaker on one dimension while remaining useful on another.

For example:

Figure: Regression Benchmark Comparison



RMSE comparison among the three individual base models — D-MPNN V2-C, CatBoost V2 (individual), and LightGBM — across all four splits. CatBoost is the strongest individual model on every split; all three models weaken substantially under COLD_DRUG.

- D-MPNN trails CatBoost on individual-model MAE across all four splits.
- D-MPNN outperforms LightGBM on three of four splits.
- D-MPNN contributes a structurally distinct molecular signal to the ensemble.

The most defensible conclusion is therefore that D-MPNN provides a biologically informed neural alternative that is complementary to the tabular models, rather than claiming it is the universally best model.

Robustness Limitation

Formal seed-repeat analysis was not included in the supplied artifacts. Specifically, the package does not provide a complete variance analysis from retraining the same architecture under multiple random seeds.

Cross-architecture comparison is therefore available, but it should not be treated as a substitute for seed-level variance estimates.

22. Trust-Layer Results

The trust-layer evidence consists of three principal analyses:

1. conformal interval coverage;
2. trust-category error separation;
3. novelty versus error relationship.

Coverage and Calibration

Empirical 90% conformal coverage is:

- **RANDOM:** 90.69%
- **COLD_COMBINATION:** 90.82%
- **COLD_CELL_LINE:** 88.49%

Mean interval width is 19.21 across all three splits.

The observed coverage is therefore close to the 90% nominal target.

COLD_CELL_LINE is mildly under-covered.

Trust-Category Separation

Across all three covered splits:

High trust → lowest MAE

Medium trust → intermediate MAE

Low trust → highest MAE

This indicates that trust categories perform genuine error-separation work rather than functioning merely as labels.

Novelty / Error Relationship

Novelty score is not consistently predictive of absolute error.

Approximate correlations are:

Split	Novelty vs. Absolute Error
RANDOM	+0.047
COLD_COMBINATION	+0.001
COLD_CELL_LINE	-0.187

The negative correlation on COLD_CELL_LINE is particularly important.

It indicates that more novel predictions do not necessarily correspond to larger errors in this dataset.

TrustSyn therefore does not present novelty as a universally validated confidence signal.

The conformal interval provides the more directly auditable uncertainty statement.

23. Explainability Results

The global CatBoost SHAP ranking places:

1. `union_target_count`
2. `Tanimoto_similarity`
3. `target_count_A`

4. `STRING_available`
5. `CellMiner_PC2`
6. `target_count_B`
7. `KEGG_overlap`

among the most influential features.

The overall pattern indicates that target-set structure is highly important to the tabular model.

Chemical similarity also contributes strongly.

Cell-line molecular state and network/pathway context provide additional information.

This suggests that TrustSyn's tabular prediction branch is not relying on a single biological signal but instead integrates multiple forms of information.

Explainability Limitation

The supplied SHAP analysis is global rather than split-specific.

Therefore, the current evidence cannot establish whether feature importance changes substantially between RANDOM and the cold-start settings.

24. Failure Modes

COLD_DRUG is the clearest and most consistent failure mode for the individual D-MPNN model.

Correlation Collapse

Pearson correlation falls from approximately:

0.543 → 0.077

between RANDOM and COLD_DRUG.

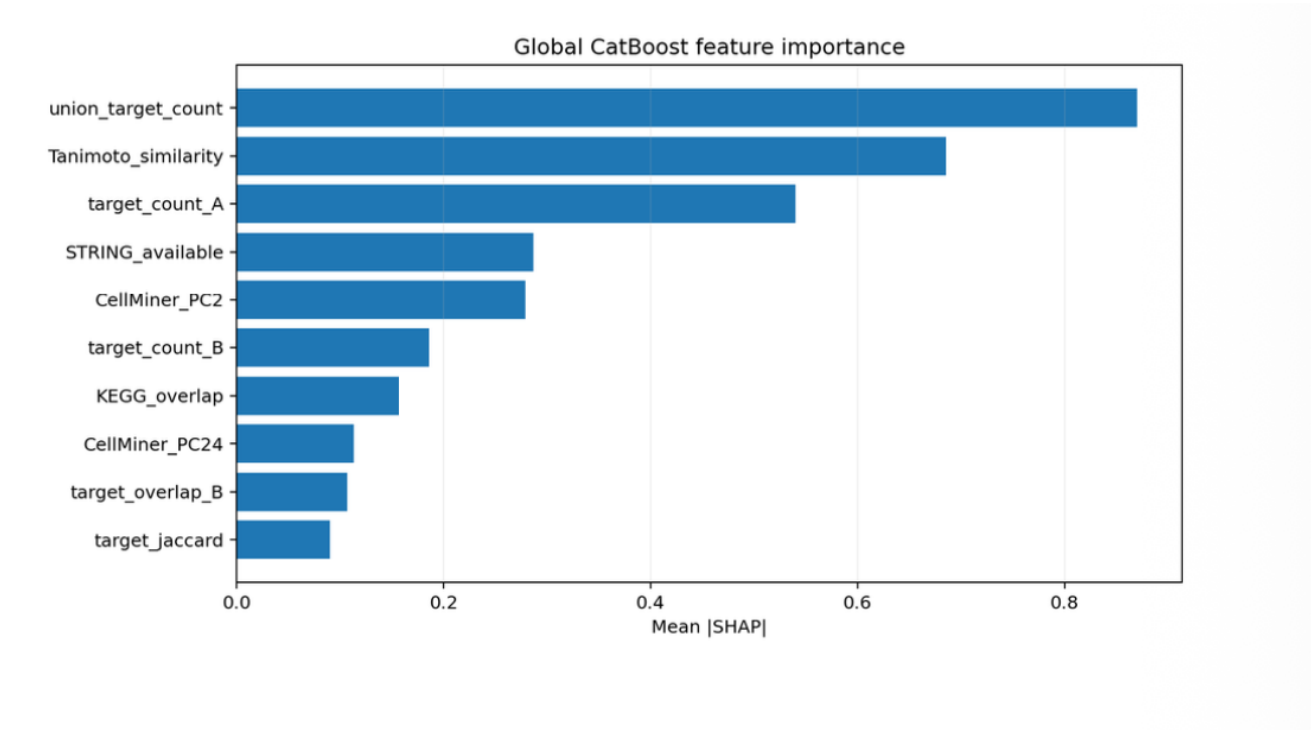
Spearman correlation falls from:

0.420 → 0.062

This indicates that predictions for genuinely unseen drugs become nearly uncorrelated with observed combo_score.

Referenced in Section 13 (Explainability) & Section 23 (Explainability Results)

Figure: Global CatBoost Feature Importance



Mean |SHAP| value for the top ten global features driving CatBoost predictions, led by union_target_count, Tanimoto_similarity, and target_count_A. Spans target-set structure, chemical similarity, network/pathway information, and cell-line molecular state.

Ranking Collapse

nDCG@100 becomes:

-0.083

and Enrichment@100 falls to:

1.57

compared with:

111.75

on RANDOM.

This means the model's top-ranked predictions on COLD_DRUG carry very little useful ranking information.

Trust-Layer Blind Spot

COLD_DRUG is outside the validated trust-layer scope.

There is therefore currently no calibrated uncertainty signal available to flag unreliable COLD_DRUG predictions.

Small Test Partition

COLD_DRUG contains only 3,132 test rows compared with approximately 29,400–29,800 rows for the other splits.

This increases the sampling uncertainty associated with individual metrics.

Secondary Failure Pattern — COLD_CELL_LINE

COLD_CELL_LINE has the largest raw MAE generalization gap.

However, unlike COLD_DRUG, its correlation and ranking performance remain meaningfully informative.

The model therefore experiences greater absolute prediction drift under unseen cell-line context but retains more useful relative ordering.

25. Discussion

Taken together, the results describe a system that performs well under interpolation and degrades in structured ways under extrapolation.

RANDOM provides the strongest results across most metric families. This is expected and should not be interpreted as evidence of deployment-level performance because RANDOM allows the model to encounter familiar drugs, combinations, and biological contexts during training.

The cold-start settings provide a more informative picture.

COLD_COMBINATION

COLD_COMBINATION produces the smallest MAE generalization gap.

This suggests that when the model has already seen the individual drugs in other combinations, it can transfer some of its learned information to a new pairing.

COLD_CELL_LINE

COLD_CELL_LINE produces the largest raw MAE gap.

This suggests that biological context is a difficult axis of generalization.

Cell-line identity is associated with many correlated biological factors, including:

- tissue background;
- mutation profile;
- gene-expression state;
- copy-number variation;
- protein state.

Compressing these factors into approximately 50 PCA components may preserve useful variation while also losing information necessary for reliable extrapolation.

COLD_DRUG

COLD_DRUG is the most severe failure when correlation and ranking are considered.

This is consistent with the challenge of learning meaningful molecular representations from only 104 drugs and then expecting those representations to generalize to entirely unseen chemical entities.

A graph encoder trained on a relatively small molecular universe cannot automatically be assumed to understand all future chemistry.

Why Stacking Matters

The individual models have different inductive biases:

- CatBoost learns from engineered tabular relationships.
- LightGBM provides an independent tabular representation.
- D-MPNN learns directly from molecular graph structure.

Their errors are therefore not necessarily identical.

Stacking can exploit this diversity by learning how to weight the different predictions.

Why the Trust Layer Matters

The trust layer provides an additional layer of interpretability and uncertainty.

Its strongest evidence is not simply the existence of an uncertainty number but the empirical fact that:

high-trust predictions have lower observed error than low-trust predictions.

However, the trust layer's scope must remain explicit.

It has been validated on:

- RANDOM;
- COLD_COMBINATION;
- COLD_CELL_LINE.

It has not been validated on COLD_DRUG.

This limitation is an important scientific result rather than something to conceal.

26. Limitations

Dataset Scope

TrustSyn is trained and evaluated on 104 drugs across 59 cell lines from NCI-ALMANAC.

It has not been tested on substantially different drug classes, targets, or biological backgrounds.

Results should therefore not automatically be assumed to transfer outside this dataset.

Distribution Shift

Performance materially degrades under all three cold-start settings.

The most severe failure is COLD_DRUG.

Cold-start metrics should therefore be treated as more representative of real extrapolative use than RANDOM performance.

Computational Versus Biological Uncertainty

The trust layer quantifies computational/statistical uncertainty.

It does not quantify:

- biological uncertainty;
- experimental variability;
- clinical uncertainty;
- therapeutic safety.

The trust layer is also not validated on COLD_DRUG.

Novelty Limitations

Novelty score does not consistently predict error.

It should not be used as a standalone confidence measure.

No Clinical Validation

No result in this report has been validated:

- in vitro;
- in vivo;
- clinically.

All results represent computational evaluation against NCI-ALMANAC.

Missing Evaluation Components

The supplied artifacts do not contain:

- classification metrics;
- complete data-efficiency curves for all four splits;
- stacked-ensemble-level MAE;
- stacked-ensemble-level Pearson;
- stacked-ensemble-level Spearman;
- stacked-ensemble ranking metrics.

Stacked-Ensemble Reporting Gap

The frozen stacking artifacts report RMSE and R^2 but do not provide a complete metric suite for the stacked ensemble.

The MAE, correlation, and ranking analyses elsewhere in this report therefore primarily describe the individual D-MPNN component.

27. Scientific Integrity / Non-Overclaiming

TrustSyn does **not** establish the following claims.

TrustSyn does not prove causal synergy

The ALMANAC combo_score represents a measured combination-activity score.

The model predicts this score.

It does not establish that a predicted drug pair is causally synergistic in a biological or clinical sense.

D-MPNN is not the best individual model

CatBoost has lower individual MAE on every split.

D-MPNN's contribution is complementary molecular information rather than superior standalone accuracy.

Trust scores are not universally validated confidence measures

The trust layer is validated only on RANDOM, COLD_COMBINATION, and COLD_CELL_LINE.

COLD_DRUG is explicitly outside its validated scope.

TrustSyn is not a clinical system

TrustSyn is a computational prioritization framework intended to support experimental follow-up.

It is not:

- a diagnostic tool;
- a treatment-selection system;
- a clinical decision-support system;
- a regulatory-approved therapeutic model.

Evaluation coverage is not uniform

Not every metric family is available for every split and every model.

This report explicitly distinguishes between available evidence and missing evidence rather than filling gaps through inference.

28. Reproducibility

The **Complete_model** package contains:

- D-MPNN evaluation;
- CatBoost model and evaluation artifacts;
- LightGBM evaluation;
- stacking tables;
- frozen results;
- dedicated COLD_DRUG stacking results;
- trust-layer CSVs;
- explainability outputs.

The frozen result directory additionally contains SHA-1 checksums for core CSV artifacts, supporting artifact-level integrity verification.

Artifact	Role
TRUSTSYN_FINAL_FROZEN_RESULTS	Frozen three-split stacking source of truth
TrustSyn_STACKED_RESULTS / COLD_DRUG	Dedicated COLD_DRUG stacked evaluation

TrustSyn_DMPNN / FINAL_EVALUATION	D-MPNN metrics, ranking, generalization, baseline, and status workbook
TrustSyn_TRUST_LAYER	Uncertainty, novelty, conformal, and trust outputs
TrustSyn_EXPLAINABILITY	CatBoost SHAP outputs
CATBOOST_TEAM_PACKAGE	CatBoost training/evaluation/provenance package

D-MPNN Output Structure

Each split is independently trained and checkpointed.

Model checkpoints

`FINAL_V2_C/{RANDOM, COLD_COMBINATION, COLD_CELL_LINE, COLD_DRUG}/checkpoints/dmpnn_final_v2_c_best.pt`

Training histories

`.../metrics/final_v2_c_training_history.csv`

Contains:

- training loss;
- validation MAE;
- validation RMSE;
- epoch timing;
- best-epoch tracking.

Training summaries

`.../metrics/final_v2_c_training_summary.json`

Contains summary information for the completed training run.

Test predictions

`.../predictions/final_v2_c_test_predictions.csv`

These are the authoritative per-split test predictions used for final evaluation.

Data efficiency

[DATA_EFFICIENCY/COLD_COMBINATION/DMPNN_COLD_COMBINATION_DATA_EFFICIENCY.csv](#)

Contains the 1/5/10/25/100% data-efficiency experiment.

Evaluation workbook

[FINAL_EVALUATION/TrustSyn_DMPNN_V2C_Final_Evaluation.xlsx](#)

Contains:

- Overall_Metrics;
- Ranking_Metrics;
- Data_Efficiency;
- Generalization_Gap;
- Baseline_Comparison;
- Robustness;
- Model_Config;
- STATUS;
- per-split prediction sheets.

Reproducibility Principles

Frozen Master

The ALMANAC master dataset is not modified downstream of its frozen release.

Frozen Splits

The canonical TrustSyn splits are not regenerated for final evaluation.

Separate Outputs

D-MPNN artifacts are stored separately from CatBoost and LightGBM outputs.

Split-Specific Training

Each canonical split receives an independent model-training process.

Best-Checkpoint Evaluation

The best validation-MAE checkpoint, rather than the final-epoch checkpoint, is used for test prediction.

Test Isolation

The test set is never used to select epochs or hyperparameters.

29. Final TrustSyn Workflow

The complete TrustSyn pipeline is:

Stage	What Happens
1. Source Data	NCI-ALMANAC → frozen master containing 294,073 observations plus CellMiner biological features
2. Split Design	Canonical RANDOM / COLD_COMBINATION / COLD_CELL_LINE / COLD_DRUG splits
3. Feature Engineering	Chemical similarity, target overlap, network/pathway features, CellMiner PCA, and molecular graphs
4. Base Models	CatBoost + LightGBM + D-MPNN V2-C
5. Stacking	Split-specific Ridge/XGBoost/dedicated meta-model
6. Trust Layer	Conformal intervals, novelty scoring, trust categories
7. Explainability	Global CatBoost SHAP analysis
8. Output	combo_score prediction + trust category + uncertainty interval + prioritization

The conceptual workflow can therefore be summarized as:

NCI-ALMANAC → Frozen Data → Leakage-Safe Splits → Biological/Chemical Features → CatBoost + LightGBM + D-MPNN → Stacking → Trust Layer → Explainability → Prioritized Drug Combinations

30. Conclusion

TrustSyn demonstrates that a biologically informed, stacked modeling system combining tabular gradient boosting with a molecular-graph neural network can generate meaningful predictions of NCI-ALMANAC drug-combination activity under multiple forms of distribution shift.

The central strength of the system is not simply its RANDOM performance.

Its more important contribution is the explicit evaluation of increasingly difficult generalization settings:

- RANDOM;
- COLD_COMBINATION;
- COLD_CELL_LINE;
- COLD_DRUG.

The results show that performance degrades systematically as the model is asked to extrapolate beyond familiar data.

RANDOM and COLD_COMBINATION provide the strongest evidence of useful predictive behavior.

COLD_CELL_LINE shows the largest raw MAE generalization gap but retains meaningful ranking information.

COLD_DRUG is the clearest limitation, with severe degradation in correlation and ranking performance.

This failure is explicitly reported rather than hidden.

The stacking framework provides complementary information by combining CatBoost, LightGBM, and D-MPNN.

The trust layer adds an additional reliability-oriented component, demonstrating useful error separation between high- and low-trust predictions and near-nominal conformal coverage across the three covered splits.

However, TrustSyn remains limited by incomplete evaluation coverage, absence of clinical or wet-lab validation, and the lack of a validated trust layer for COLD_DRUG.

The most immediate future improvements are therefore:

1. extending the trust layer to COLD_DRUG;
2. completing ranking and classification evaluation;
3. completing data-efficiency curves for all four splits;
4. reporting the complete metric suite directly for the stacked ensemble;

5. increasing chemical diversity and training scale for stronger unseen-drug generalization.

The overall conclusion is therefore not that TrustSyn has solved drug-combination prediction.

Rather, TrustSyn establishes a reproducible, biologically informed framework that can identify promising computational predictions while explicitly communicating where its predictions are reliable and where additional experimental evidence is required.

31. AI Assistance Disclosure

ChatGPT (OpenAI) was used throughout the project for:

- code drafting;
- debugging;
- data-analysis workflow support;
- technical writing;
- documentation;
- presentation development;
- critical review of implementation logic.

This report was compiled and restructured with assistance from Claude (Anthropic), drawing directly on the team's frozen artifact package and standalone model documentation without introducing new numerical results beyond those contained in the source artifacts.

The team retained responsibility for:

- scientific reasoning;
- methodological decisions;
- model development;
- model evaluation;
- interpretation;
- metric verification;
- final submission decisions.

AI-generated suggestions were reviewed against project artifacts before inclusion.

32. References

1. **NCI-ALMANAC** — National Cancer Institute ALMANAC (A Large Matrix of Anti-Neoplastic Agent Combinations) drug-combination screening dataset.
 2. **CellMiner** — National Cancer Institute CellMiner database of molecular profiling data, including RNA, CNV, mutation, and protein information for the NCI-60 cell-line panel.
 3. **STRING** — Protein–protein interaction network database used for target/network-overlap features.
 4. **KEGG** — Kyoto Encyclopedia of Genes and Genomes, used for pathway-overlap features.
 5. **D-MPNN** — Directed Message Passing Neural Network architecture for molecular property prediction, adapted here to the two-drug combination setting.
 6. **CatBoost** — Gradient-boosting framework used as the primary engineered-feature tabular model.
 7. **LightGBM** — Gradient-boosting framework used as a secondary tabular baseline.
 8. **XGBoost** — Gradient-boosting framework used as a stacking meta-model for selected evaluation settings.
 9. **Ridge Regression** — Linear regularized regression used as the selected stacking meta-model for RANDOM.
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Final Source-of-Truth Statement

All numerical claims in this report should be interpreted according to the artifact provenance described above.

Where an earlier model-development document and the frozen evaluation package contain conflicting numerical values, the frozen package takes precedence.

Where an evaluation component is missing, incomplete, or outside the validated scope of TrustSyn, this report explicitly states that limitation rather than estimating, interpolating, or presenting an unsupported result as final.

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